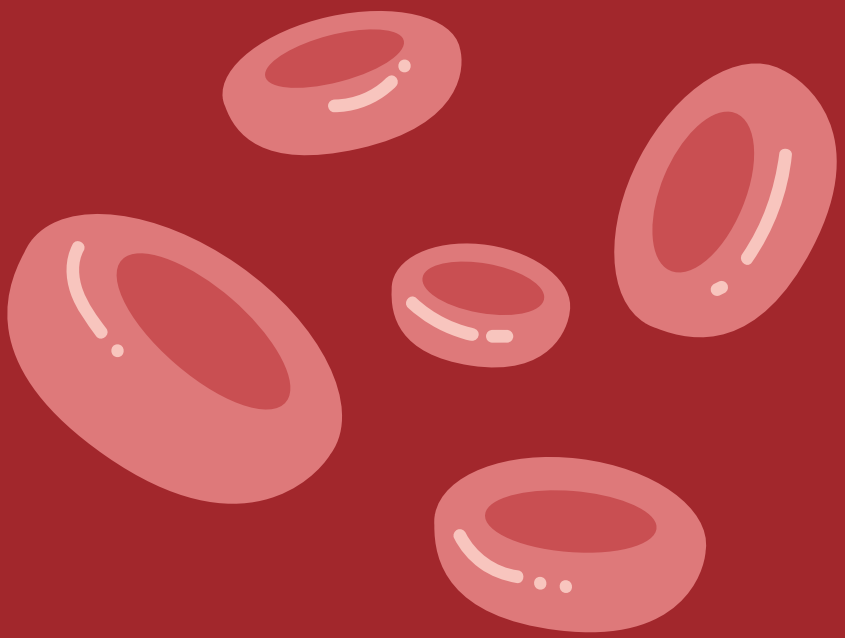
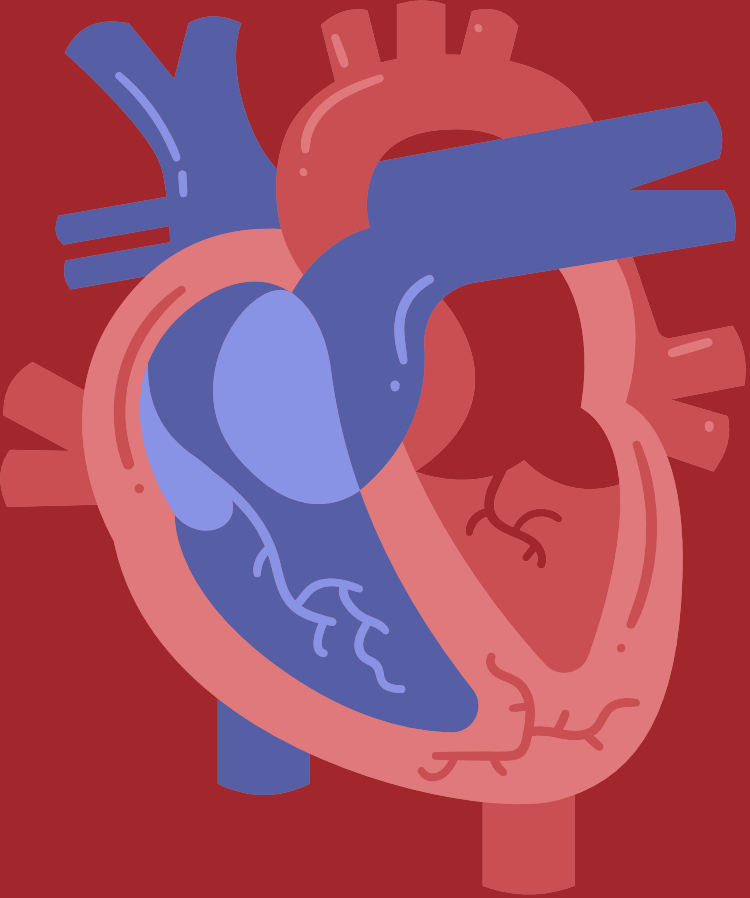
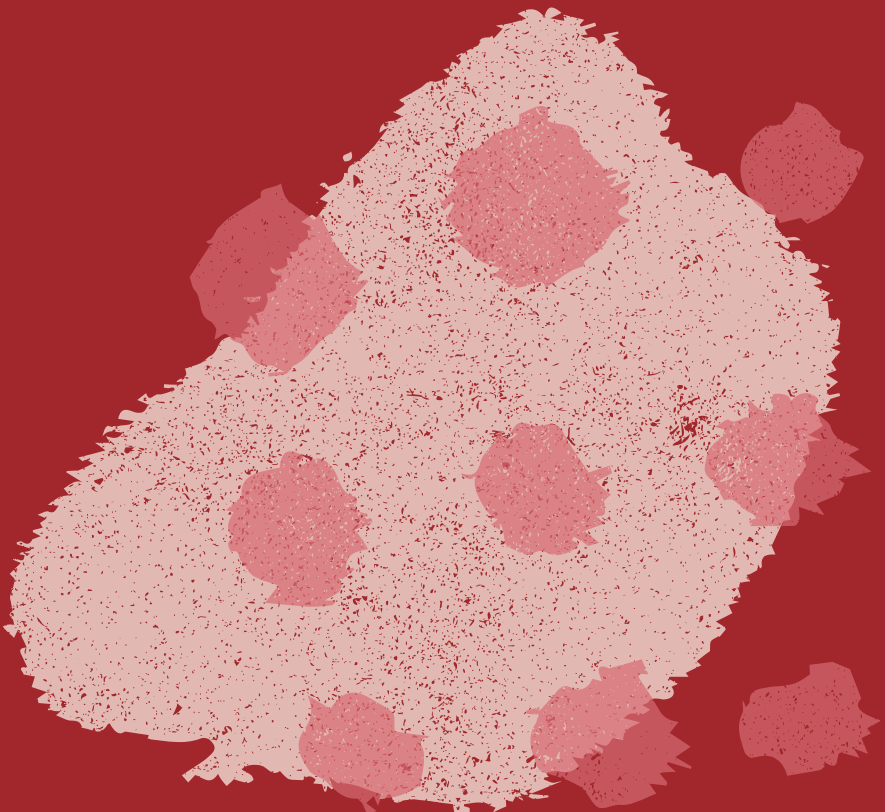




HEART ATTACK PREDICTION

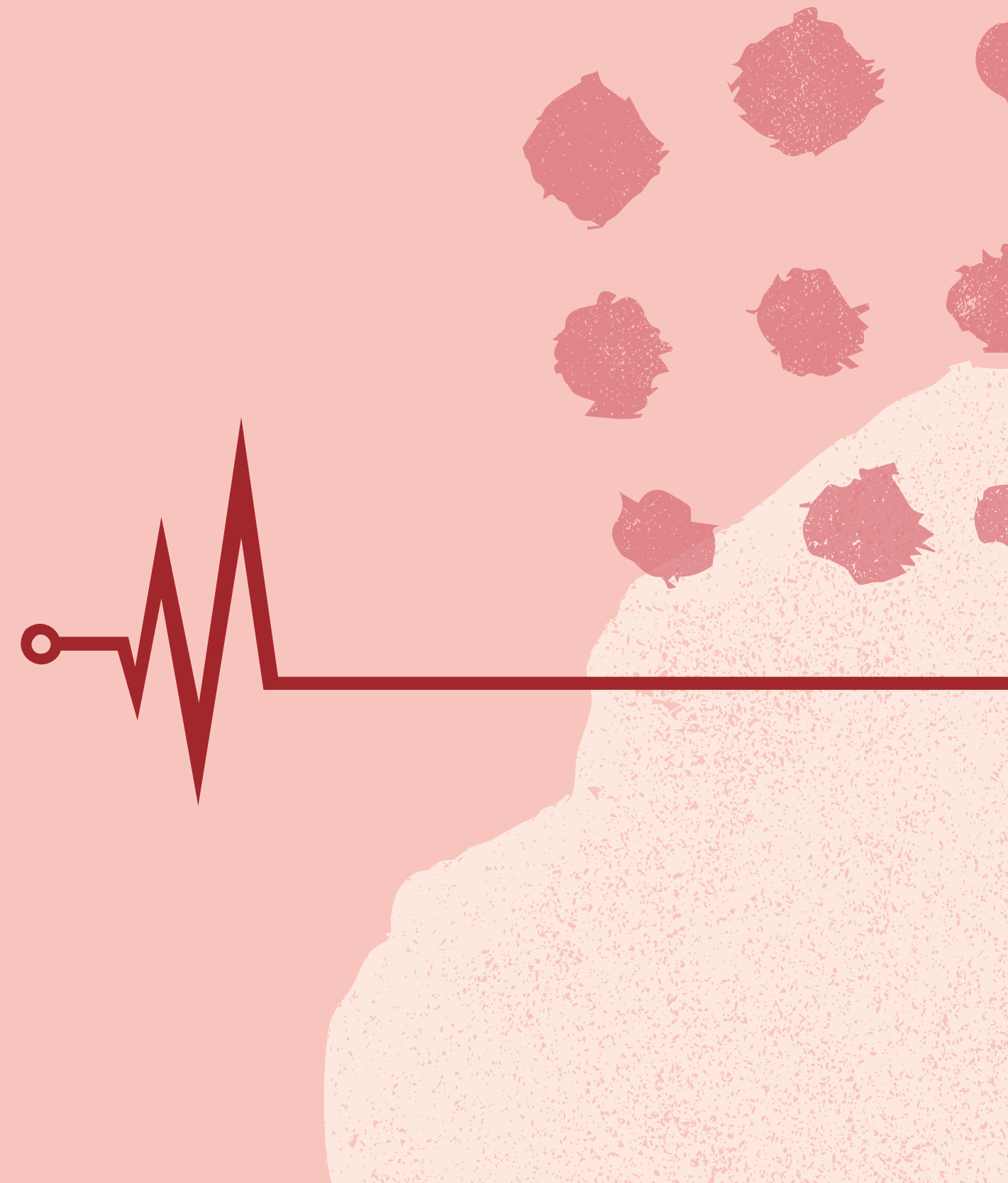
Comp 4980: Final Project

Begimai Alisherova, Raiyan Mokhammad



PROBLEM STATEMENT

Heart disease, including heart attacks, remains one of the leading causes of death globally. Predicting and preventing heart attacks on an individual level can have a significant impact on public health by reducing mortality rates and improving overall population health.





DATASET

Source: Kaggle (~250k records)

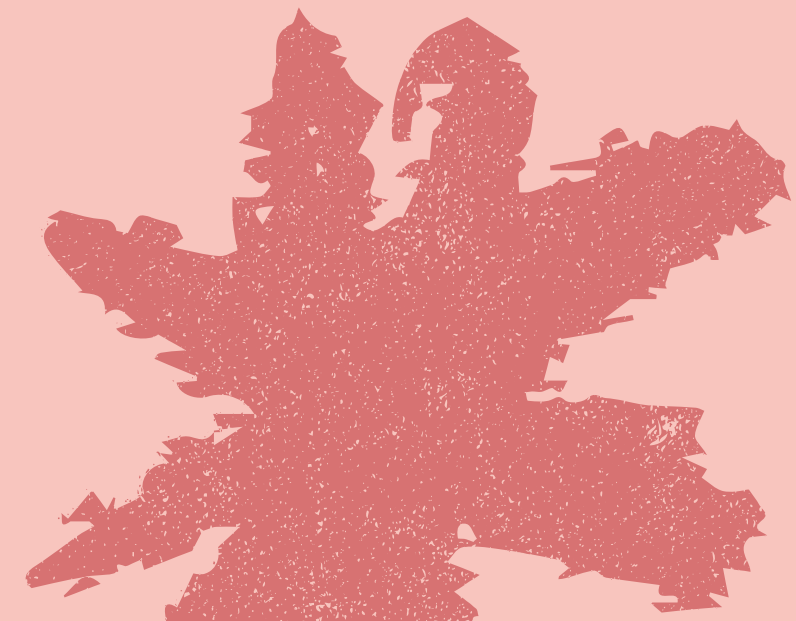
Key features:

- BMI
- Age, sex
- Lifestyle (Physical activity, Sleep hours, Mental health days)
- Habits (Drinking, Smoking)
- Health history (Diabetes, Arthritis)

ALGORITHM

Decision Tree - ID3

- Created a prediction model in Power BI
- Encoded data for ID3



MINING ALGORITHM IMPLEMENTATION

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1		Sex	GeneralHealth	PhysicalHealthDays	MentalHealthDays	SleepHours	HadStroke	HadSkinCancer	HadDepressiveDisorder	HadDiabetes	SmokerStatus	AgeCategory	BMI	AlcoholDrinkers	HadHeartAttack
2	Sex	1													
3	GeneralHealth	-0.01811	1												
4	PhysicalHealthDays	0.044081	-0.48909	1											
5	MentalHealthDays	0.091705	-0.26854	0.3068	1										
6	SleepHours	0.015793	0.068927	-0.05606	-0.1301	1									
7	HadStroke	-0.00181	-0.15891	0.128731	0.042087	0.008332	1								
8	HadSkinCancer	0.000535	-0.02646	0.030154	-0.03954	0.044071	0.039477	1							
9	HadDepressiveDisorder	0.143076	-0.20988	0.198689	0.425058	-0.04862	0.04297	-0.01457	1						
10	HadDiabetes	-0.02246	-0.27313	0.157089	0.033788	0.001038	0.107213	0.027936	0.048499	1					
11	SmokerStatus	-0.04327	-0.17329	0.120174	0.125453	-0.05051	0.058888	-0.01398	0.113325	0.035316	1				
12	AgeCategory	0.055689	-0.13014	0.089503	-0.1879	0.119721	0.130103	0.237745	-0.11227	0.20238	0.046151	1			
13	BMI	-0.00778	-0.24344	0.116905	0.082182	-0.05475	0.019942	-0.04639	0.105716	0.209936	-0.00327	-0.01719	1		
14	AlcoholDrinkers	-0.09382	0.178229	-0.12937	-0.02599	-0.00354	-0.07113	0.011512	-0.03116	-0.15061	-0.00306	-0.11879	-0.07087	1	
15	HadHeartAttack	-0.07332	-0.18568	0.13342	0.025892	0.003631	0.177137	0.049408	0.023706	0.141988	0.077336	0.167879	0.030413	-0.07418	1

AF	AG	AH
AgeCategory ▾	CAT Age ▾	Age2 ▾
Age 65 to 69	Senior	60+
Age 70 to 74	Senior	70+
Age 75 to 79	Senior	70+
Age 80 or older	Senior	80+
Age 80 or older	Senior	80+
Age 50 to 54	Middle-Aged	50+
Age 80 or older	Senior	80+
Age 75 to 79	Senior	70+
Age 40 to 44	Middle-Aged	40+
Age 75 to 79	Senior	70+
Age 80 or older	Senior	80+

=IFS(OR(AF3="Age 18 to 24", AF3="Age 25 to 29"), "18+", OR(AF3="Age 30 to 34", AF3="Age 35 to 39"), "30+", OR(AF3="Age 40 to 44", AF3="Age 45 to 49"), "40+", OR(AF3="Age 50 to 54", AF3="Age 55 to 59"), "50+", OR(AF3="Age 60 to 64", AF3="Age 65 to 69"), "60+", OR(AF3="Age 70 to 74", AF3="Age 75 to 79"), "70+", AF3="Age 80 or older", "80+")

V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP
Difficult ▾	Difficult ▾	Difficult ▾	Difficult ▾	Difficult ▾	Smoker ▾	Smoking ▾	ECigare ▾	ChestSc ▾	RaceEth ▾	AgeCat ▾	CAT Age ▾	Age2 ▾	Height ▾	Weight ▾	BMI ▾	BMI2 ▾	Alcohol ▾	HIVTest ▾	FluVaxL ▾	Pneun
No	No	No	No	No	Former sm	Yes	Never used	No	White only	Age 65 to 6	Senior	60+	1.6	71.67	27.99	Overweigh	No	No	Yes	Yes
No	No	No	No	No	Former sm	Yes	Never used	No	White only	Age 70 to 7	Senior	70+	1.78	95.25	30.13	Obesity	No	No	Yes	Yes
No	Yes	No	No	No	Former sm	Yes	Never used	Yes	White only	Age 75 to 7	Senior	70+	1.85	108.86	31.66	Obesity	Yes	No	No	Yes

MINING ALGORITHM IMPLEMENTATION

```
feature_names = (['Age', 'BMI', 'GeneralHealth', 'Sex', 'HadDepressiveDisorder',  
'Exercises', 'Alcohol', 'HadStroke',  
'MentalHealthDays', 'SleepHours', 'SmokerStatus'])
```

```
# SPLITTING DATASET INTO TRAINING AND TESTING SET  
X1_train, X1_test, y1_train, y1_test = train_test_split(X1, y1, test_size=0.2,  
random_state=42, stratify = y1)
```

```
# Model Accuracy, how often is the classifier correct?  
print("Accuracy:", accuracy_score(y_test, y_pred))  
print("Confusion Matrix:", conf_matrix)  
print("\nClassification Report:")  
print(classification_rep)
```

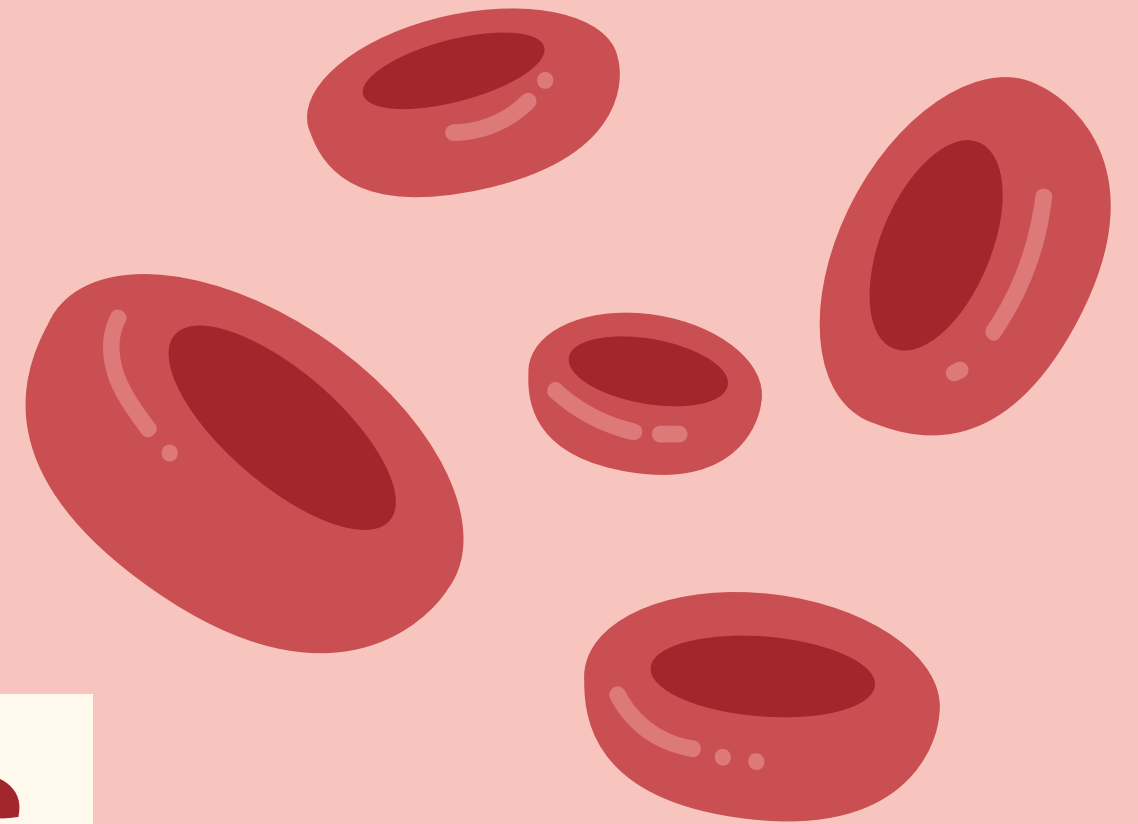
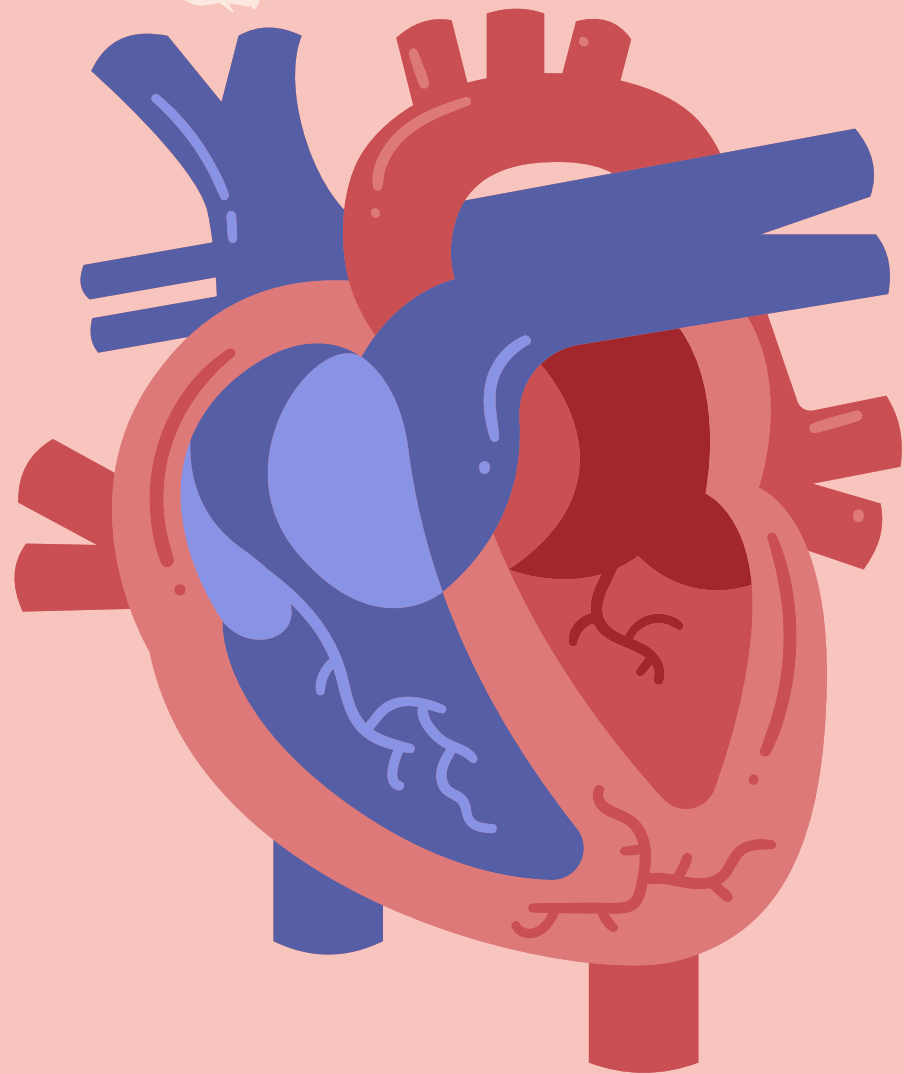
Accuracy: 0.9450055888629204

Confusion Matrix: [[46443 75]
[2631 56]]

Classification Report:

	precision	recall	f1-score	support
0	0.95	1.00	0.97	46518
1	0.43	0.02	0.04	2687
accuracy			0.95	49205
macro avg	0.69	0.51	0.51	49205
weighted avg	0.92	0.95	0.92	49205

POWER BI VISUALS



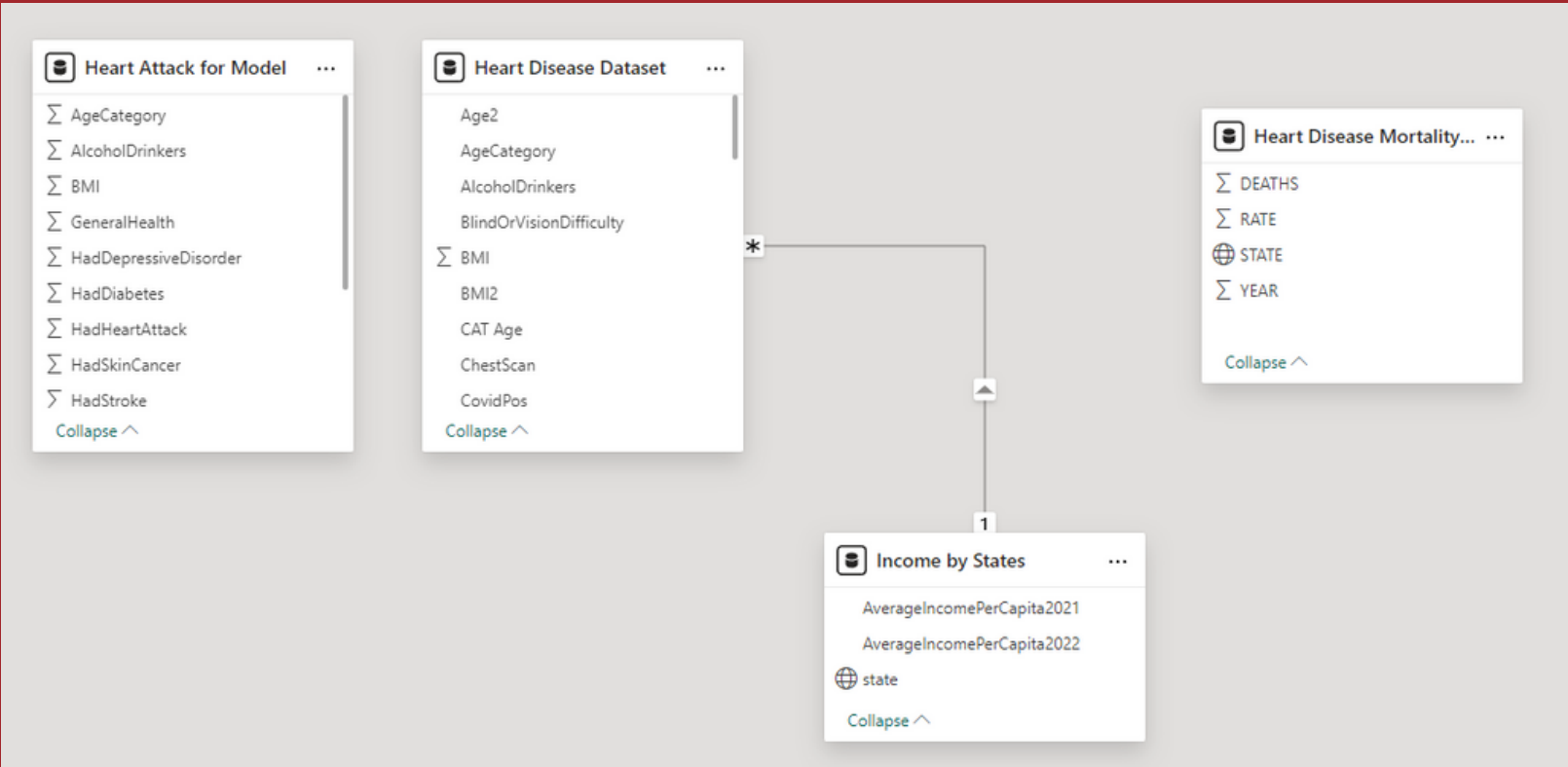
DATASET IMPORT

✕ ✓		✓									
State	Sex	GeneralHealth	PhysicalHealthDays	Physical2	MentalHealthDays	LastCheckupTime	PhysicalActivities	SleepHours	RemovedTeeth		
Alabama	Female	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Alaska	Male	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Alaska	Female	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Alaska	Male	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Arizona	Female	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Arizona	Male	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Arizona	Female	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Arizona	Female	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Arkansas	Male	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
California	Male	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Colorado	Female	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Colorado	Female	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Colorado	Female	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Connecticut	Male	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Connecticut	Female	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Connecticut	Female	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Connecticut	Female	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Connecticut	Male	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Connecticut	Male	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Delaware	Male	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Delaware	Female	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
District of Columbia	Female	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Florida	Female	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Florida	Male	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Florida	Female	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Florida	Male	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		
Florida	Female	Very good	0	Low	0	Within past year (anytime less than 12 months ago)	Yes	7	None of them		

PREPROCESSING IN POWER BI

- Creating relationships between tables
- Manipulating columns and their values in Power Query

As we do not have any dates, we did not use Hierarchies or Date tables.



Name

state

Format

Text

Summarization

Don't summarize

Sort

Sort by column

Data groups

Groups

Manage relationships

Relationships

New column

Calculations

Data type

Text

Formatting

\$ %

Auto

Properties

Data category

State or Province

Structure

×

✓

state	AverageIncomePerCapita2022	AverageIncomePerCapita2021
Alabama	50637	49671
Alaska	68919	65677
Arizona	56667	55574
Arkansas	51787	50588
California	77339	76800
Colorado	74167	70715
Connecticut	84972	82885
Delaware	61387	59846
District of Columbia	96728	96659
Florida	63597	62136
Georgia	57129	55846
Hawaii	61175	60711

Data

Search

Heart Attack for Model

Heart Disease Dataset

Heart Disease Mortality Rate by S...

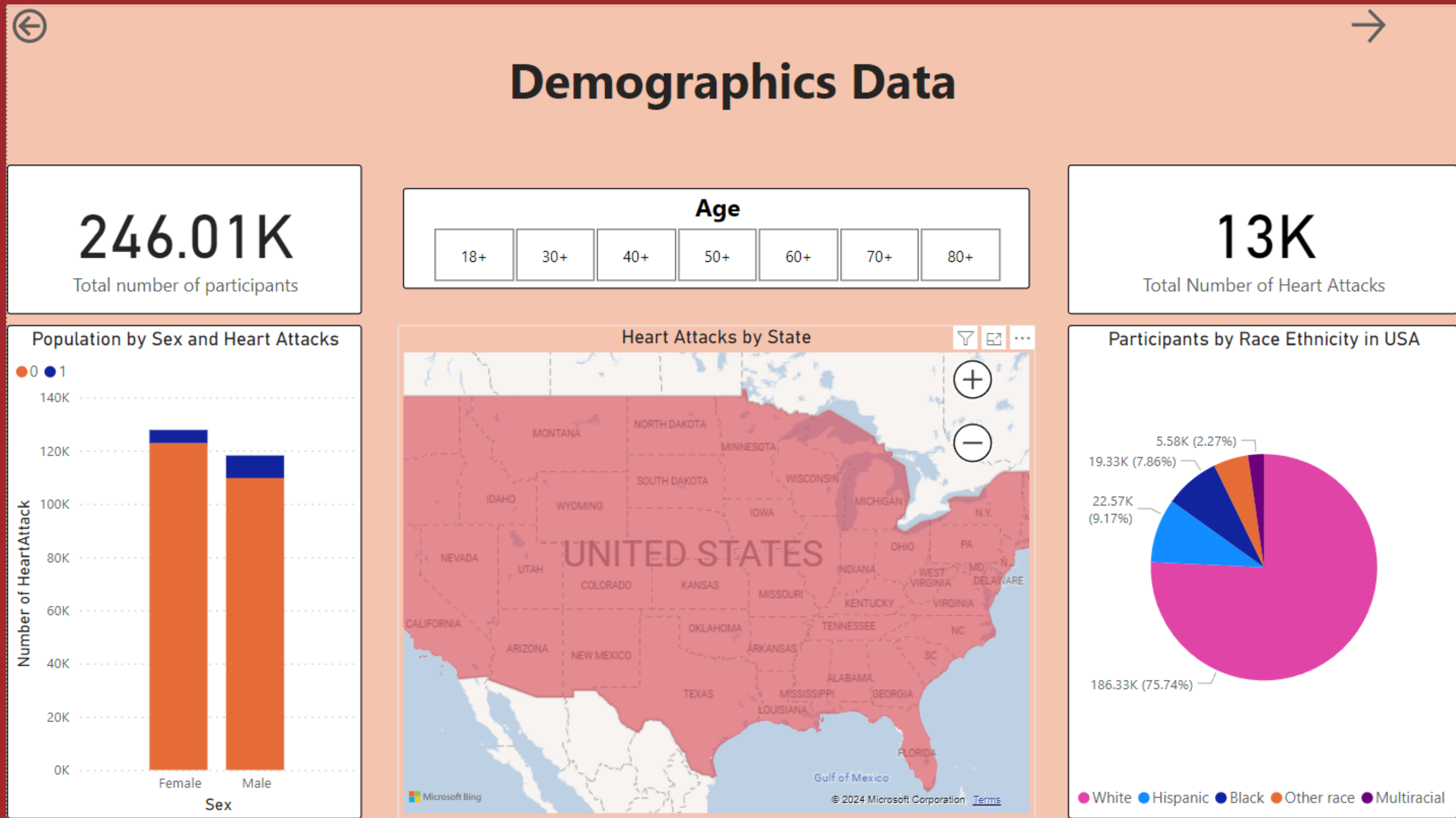
Income by States

AverageIncomePerCapita2021

AverageIncomePerCapita2022

state

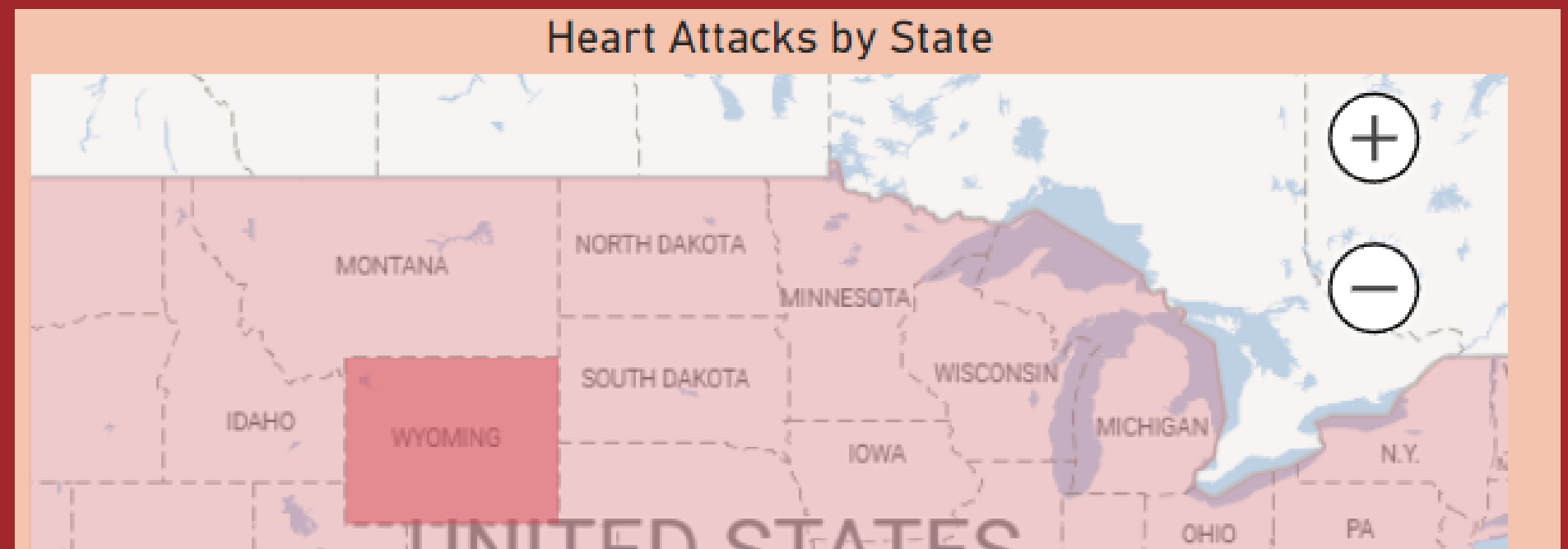
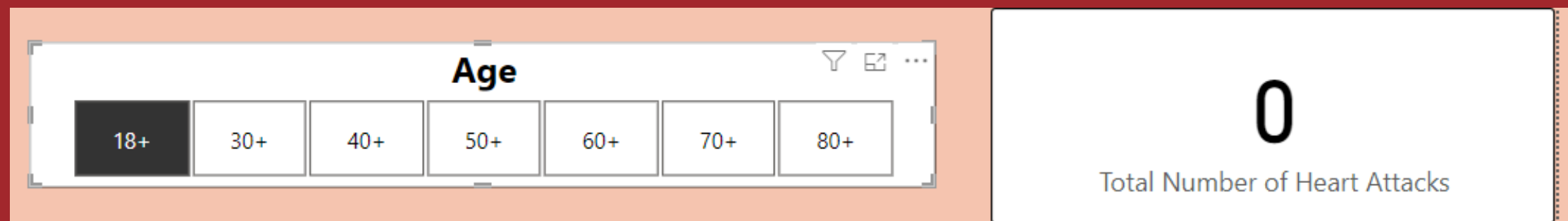
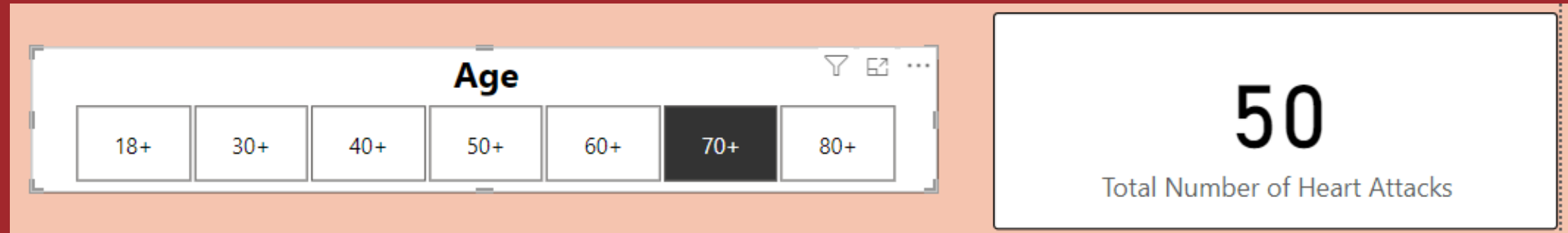
VISUALS: OVERVIEW



VISUALS: OVERVIEW

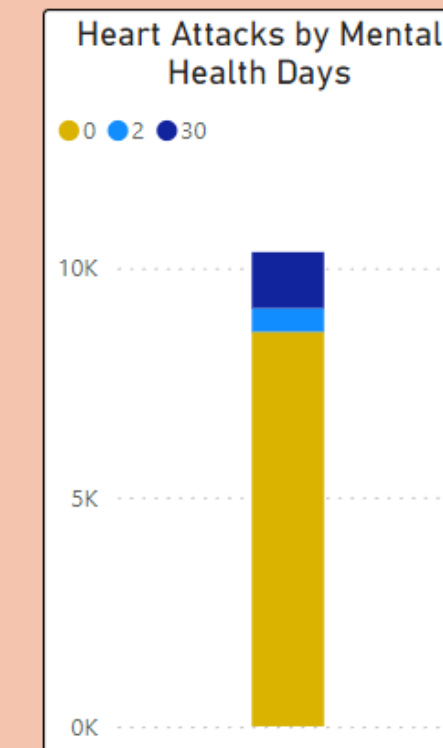
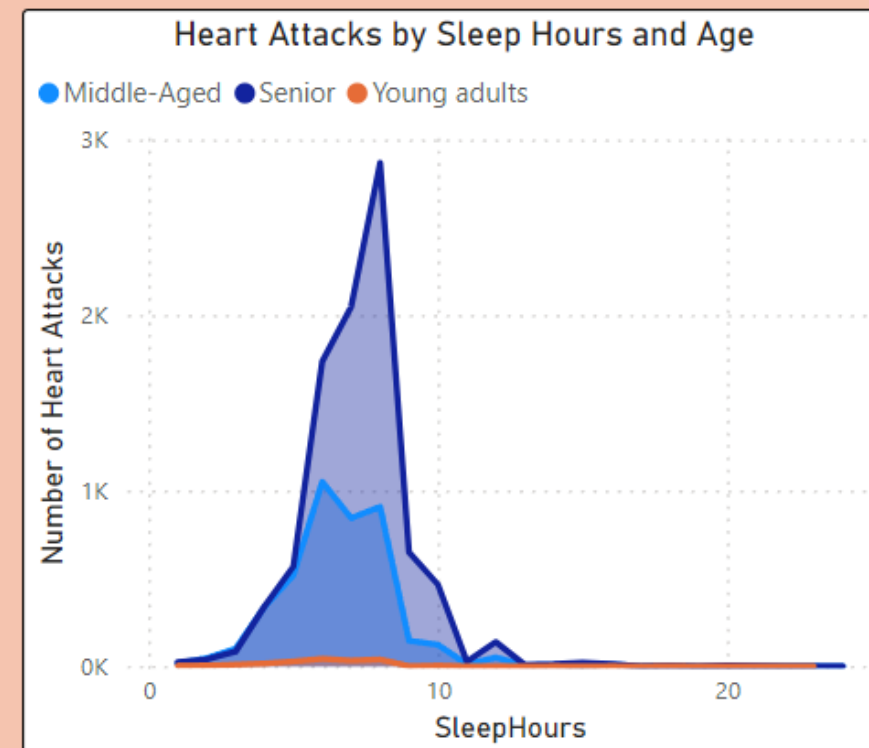
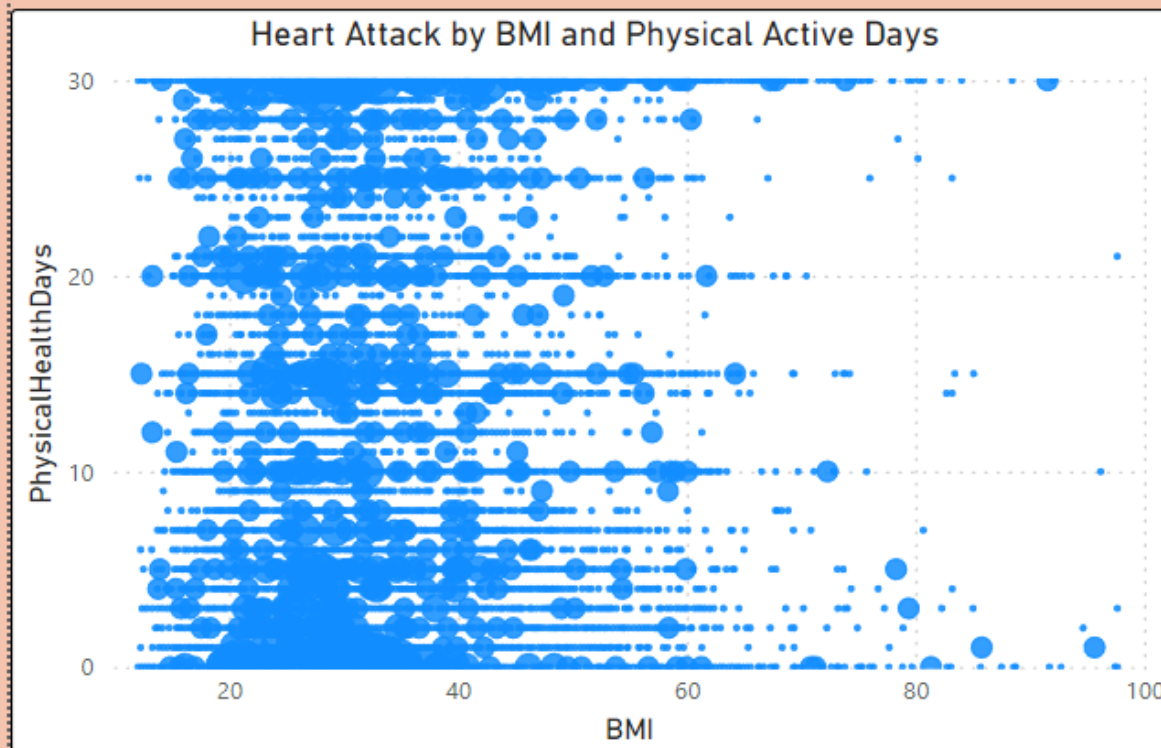
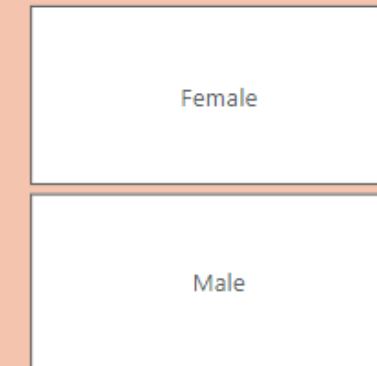
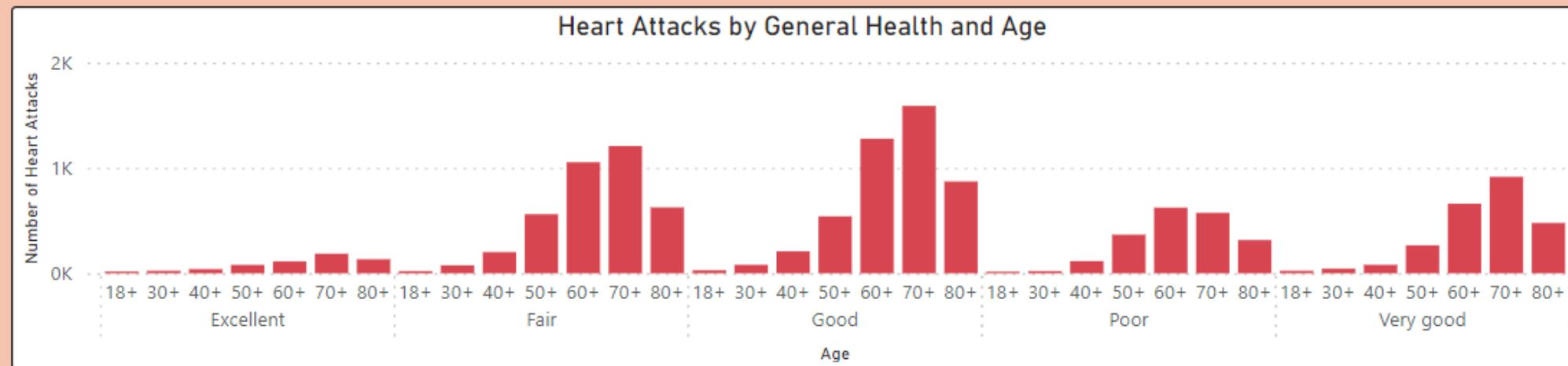
Visuals used:

- Slicer
- Cards
- Stacked Column chart
- Filled Map
- Pie Chart
- Button
- Bookmark



VISUALS: HEALTH

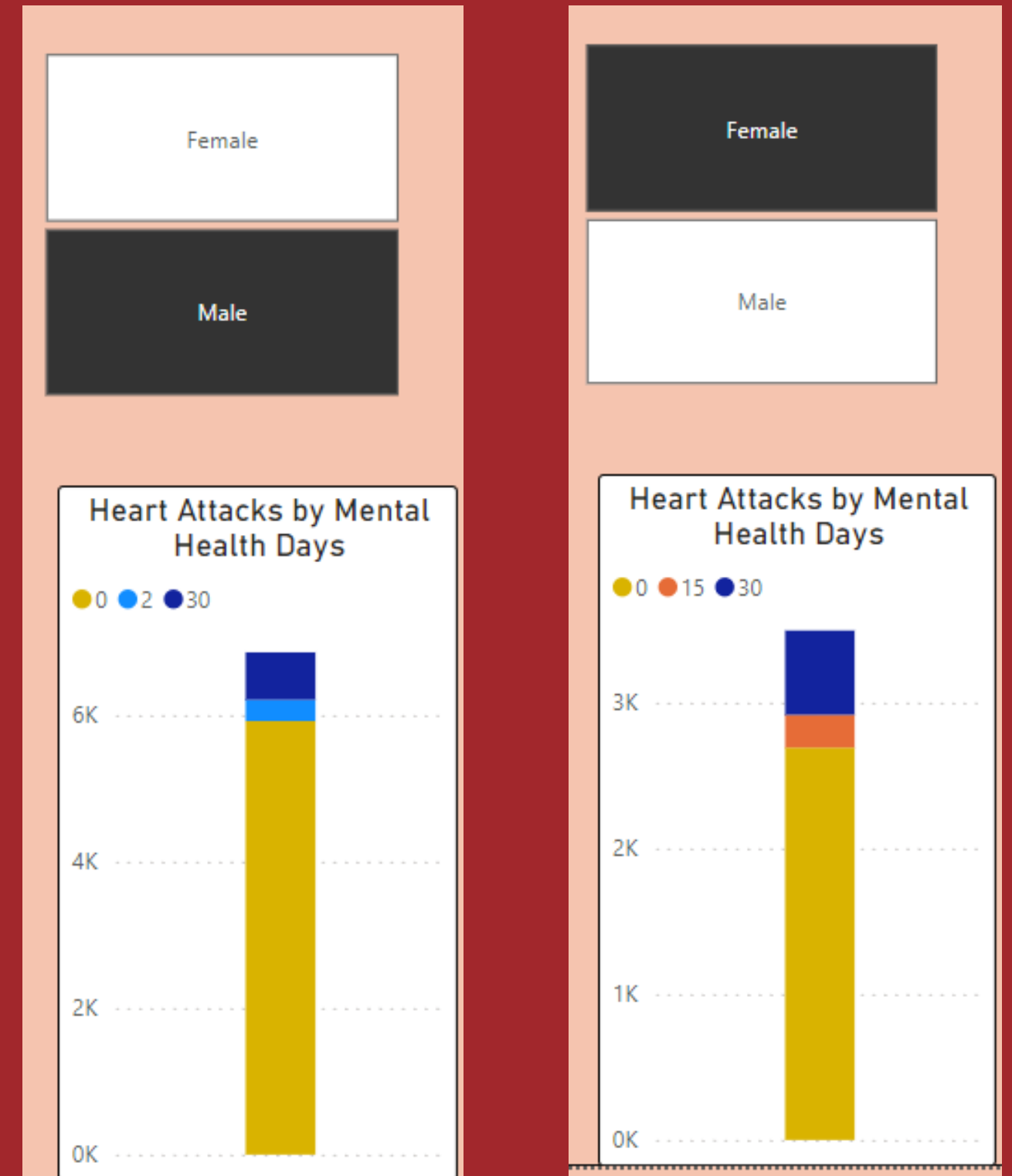
Health Data



VISUALS: HEALTH

Visuals used:

- Slicer
- Line and Stacked Column Chart
- Scatter Chart
- Area Chart
- Stacked Column chart
- Button
- Bookmark

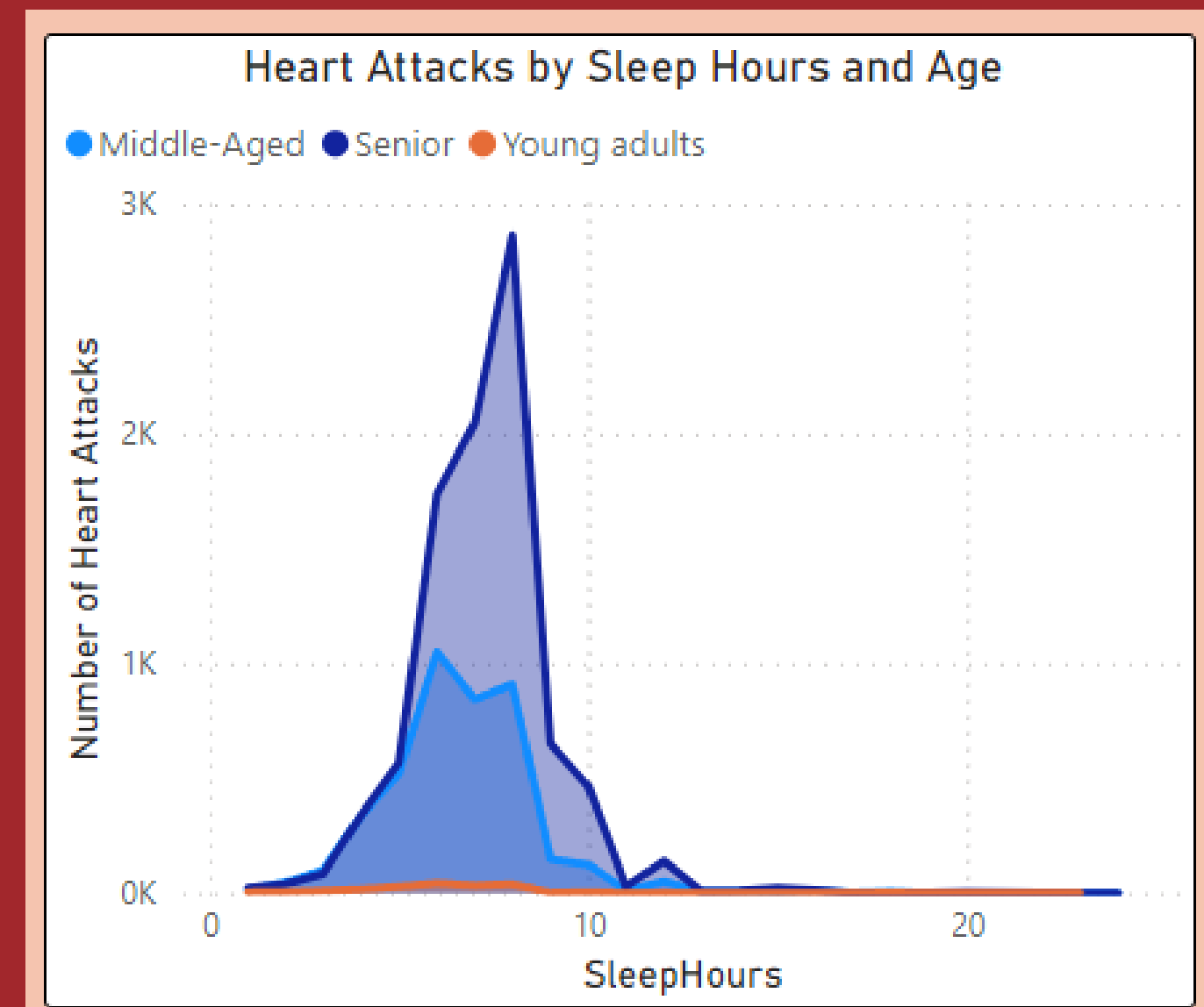


VISUALS: HEALTH

Senior (65+) have the most number of heart attacks.

Middle-aged adults (35–65) who sleep on average 6 hours a day have the highest heart attack rates among their age group.

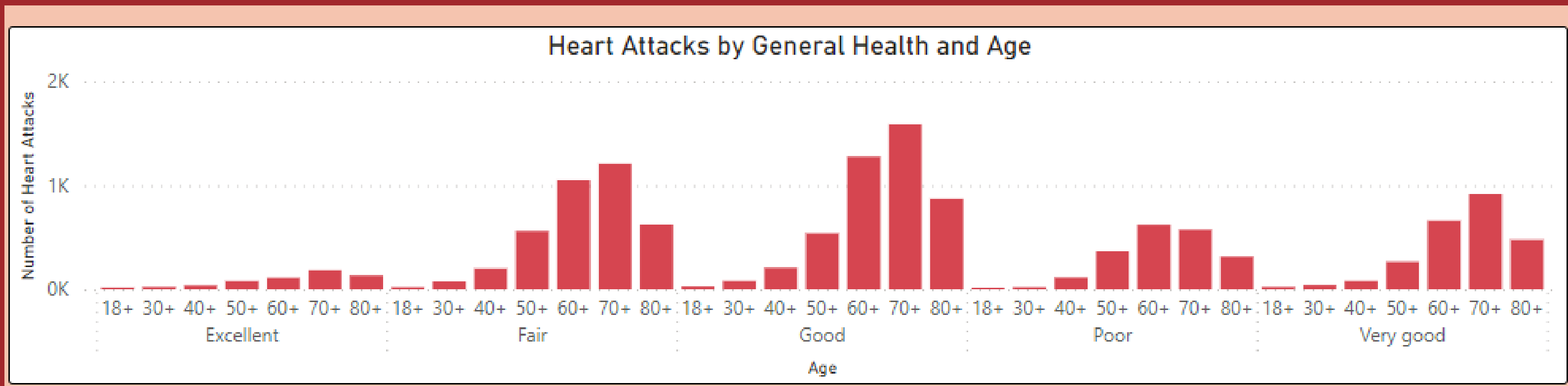
Result: sleep well!!!



VISUALS: HEALTH

Age is the most crucial attribute in determining heart attack in our dataset.

Having an excellent general health reduces the probability of heart attack.

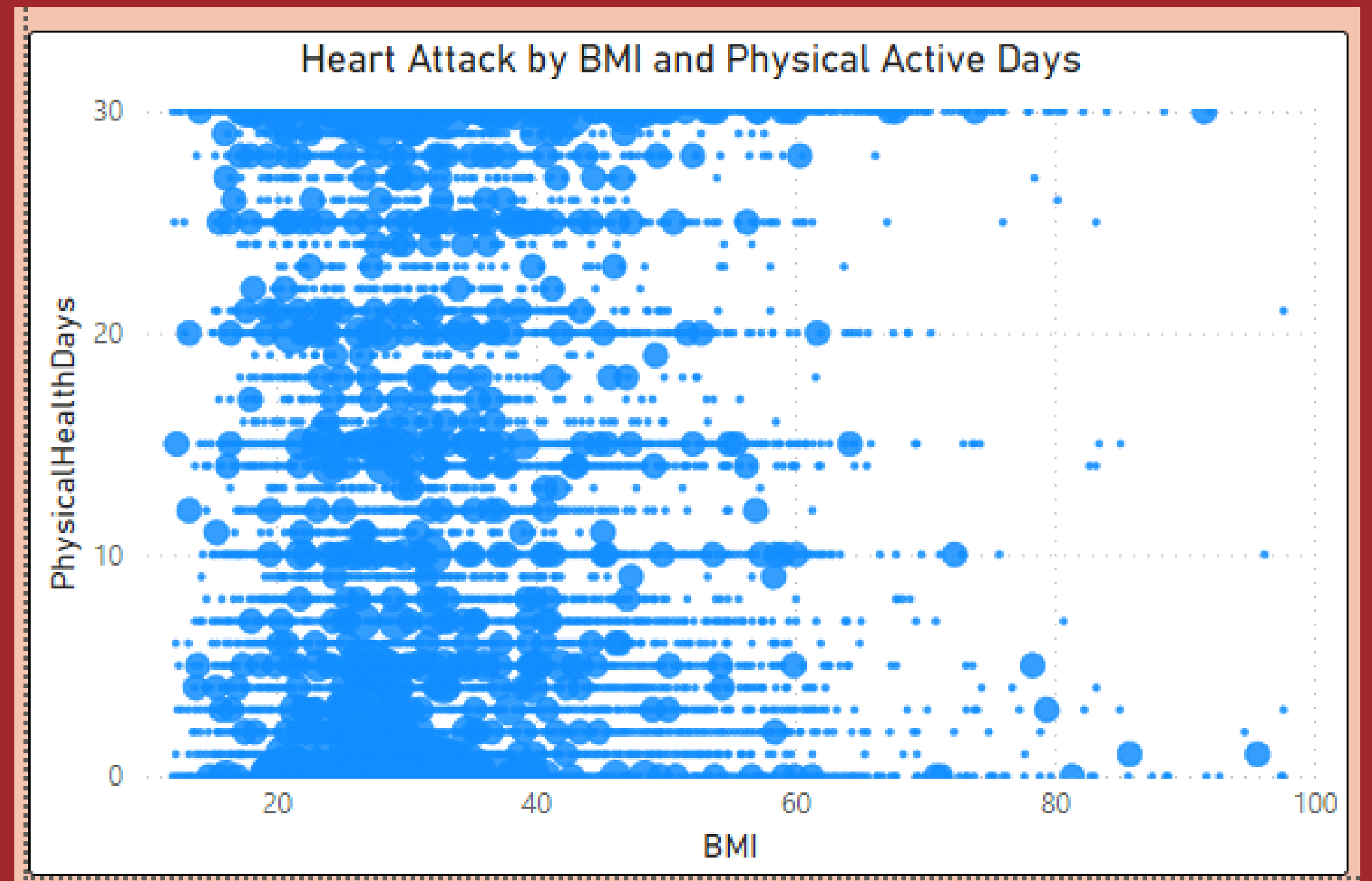


VISUALS: HEALTH

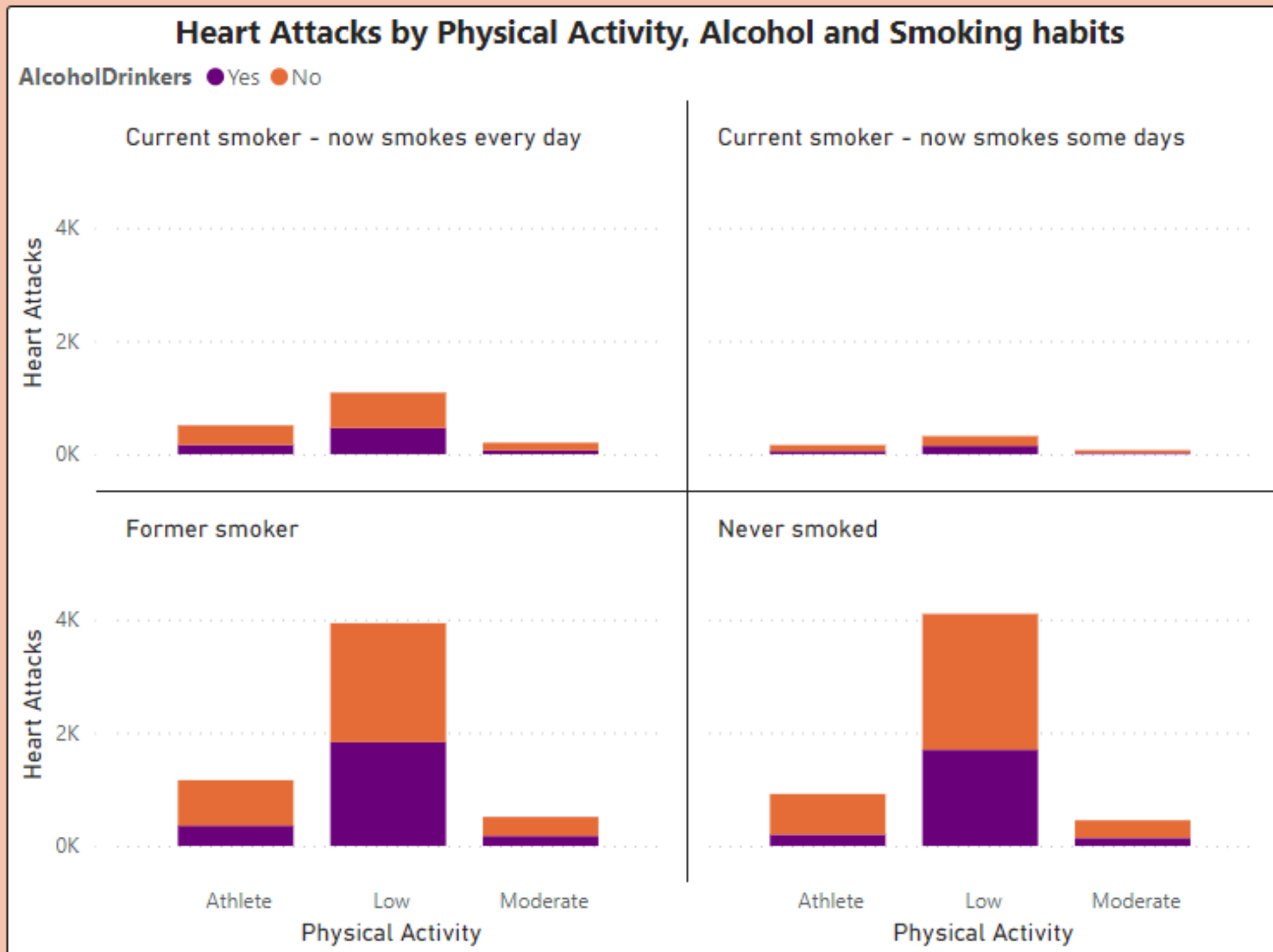
Small bubbles – all participants.
Big bubbles – participants with heart attack.

Healthy BMI – greater than 25.

Results: High BMI and low physical activity lead to higher chances of heart attack!

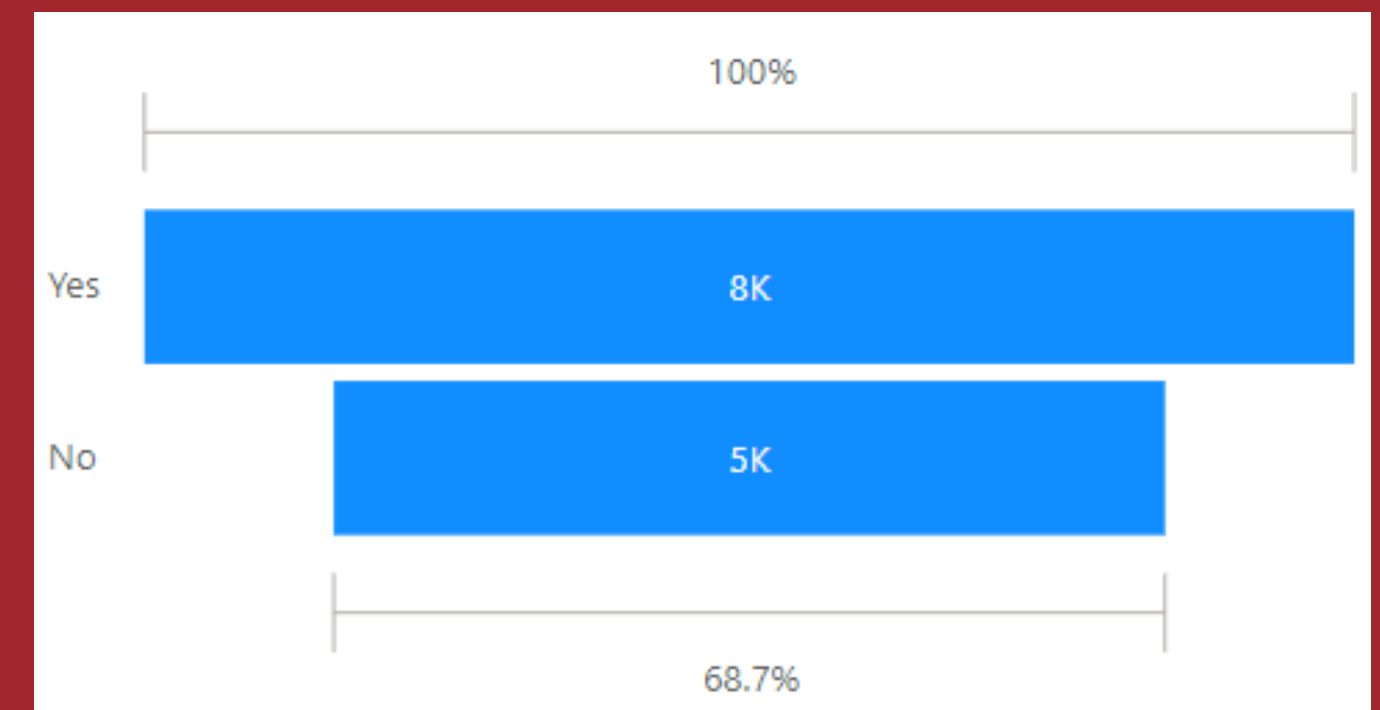


VISUALS: HEALTH



8k people who have heart attack
have smoking experience

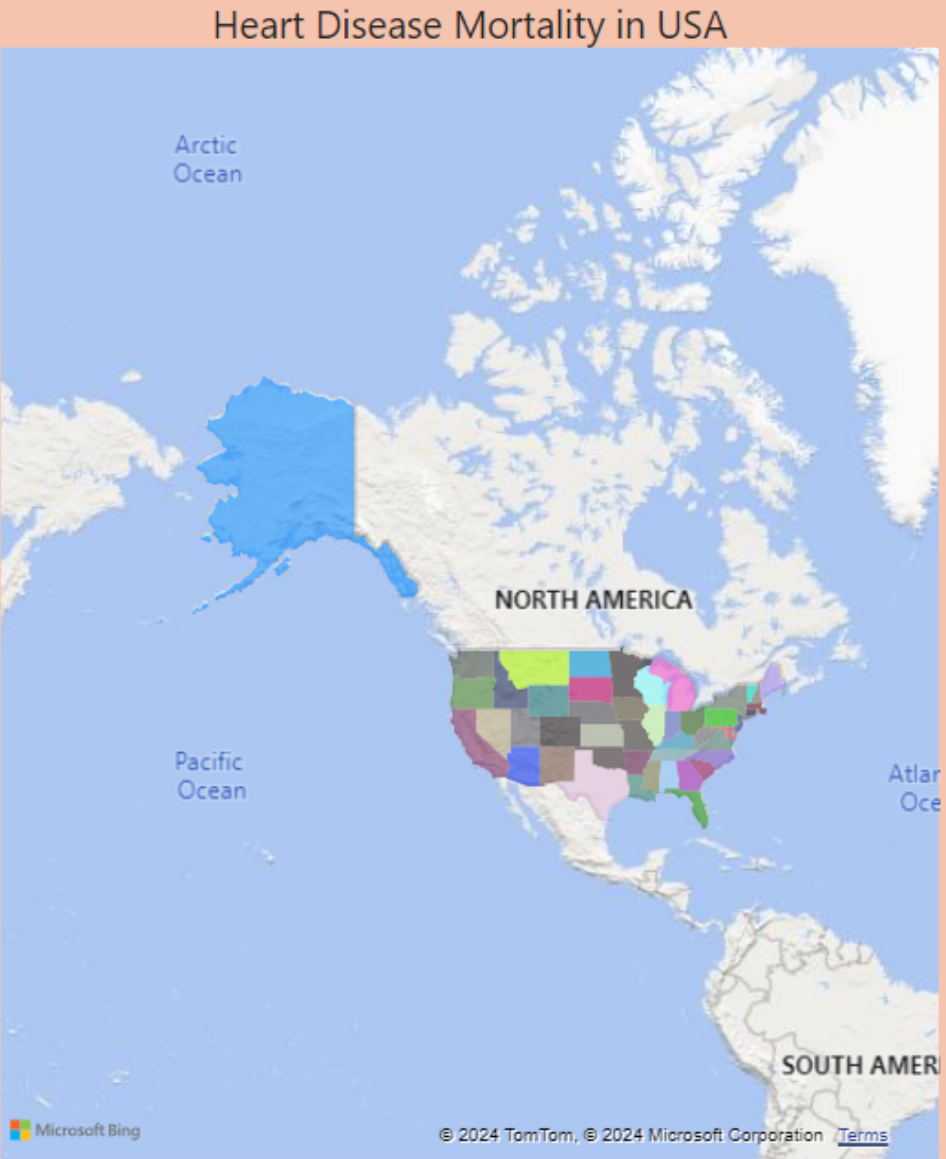
at the same time there is no
direct relationship between heart
attacks and drinking habits



VISUALS: US GENERAL

Overview of US

State	AvgIncomePerCapita2021	Number of Heart Attacks
		274
Alabama	49,671.00	124
Alaska	65,677.00	165
Arizona	55,574.00	359
Arkansas	50,588.00	244
California	76,800.00	214
Colorado	70,715.00	210
Connecticut	82,885.00	230
Delaware	59,846.00	108
District of Columbia	96,659.00	59
Florida	62,136.00	557
Georgia	55,846.00	276
Hawaii	60,711.00	249
Idaho	52,276.00	166
Illinois	67,165.00	108
Indiana	56,435.00	339
Iowa	57,080.00	271
Kansas	58,857.00	331
Kentucky	51,298.00	158
Louisiana	54,181.00	179
Maine	58,272.00	425
Maryland	69,710.00	496
Massachusetts	83,593.00	247
Michigan	56,569.00	301
Minnesota	66,232.00	396
Mississippi	45,887.00	133
Total		13435



Mortality from heart diseases in US, 2021

6M

Average Mortality Rate in US, 2021

172.38

Average Income in US, 2021

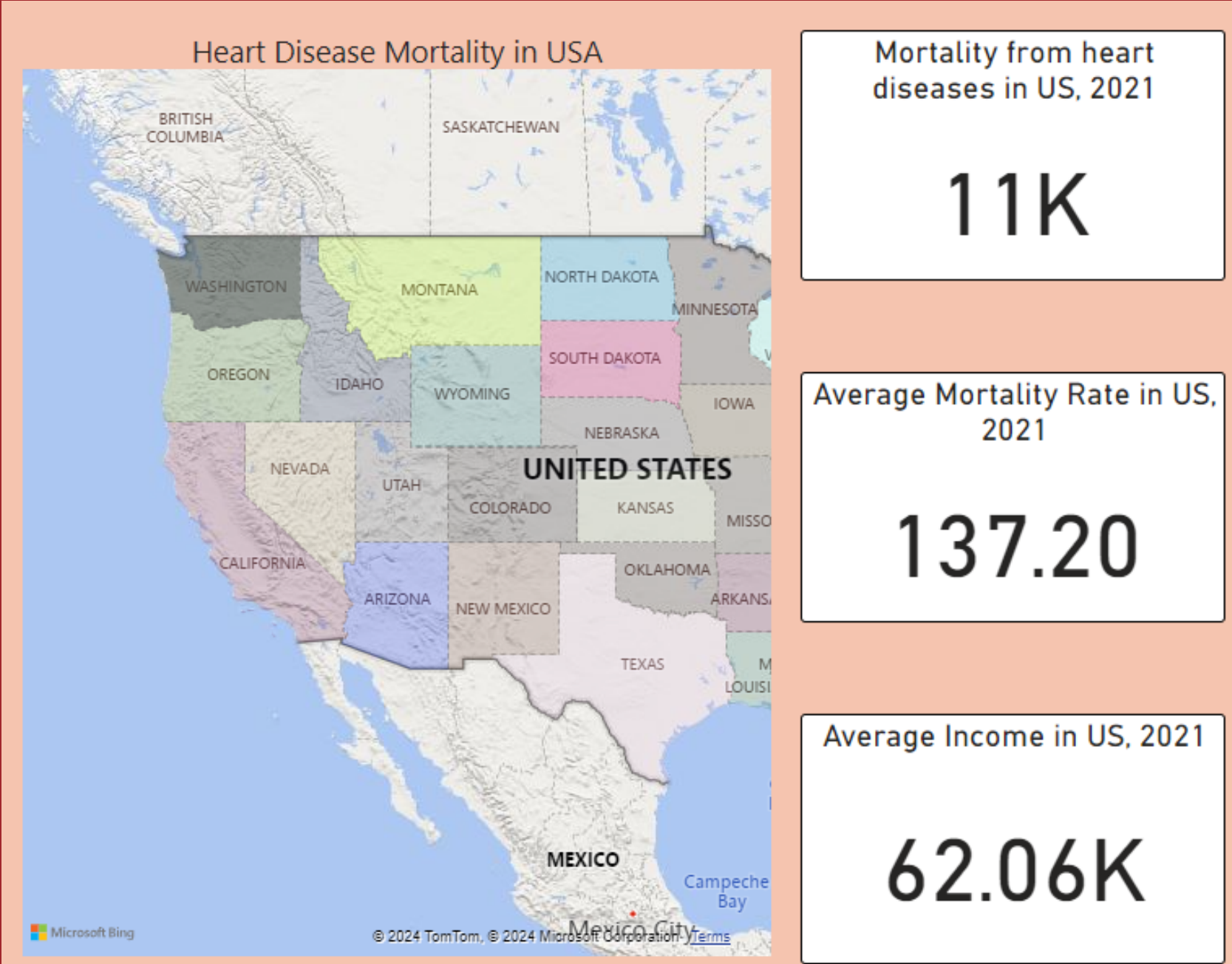
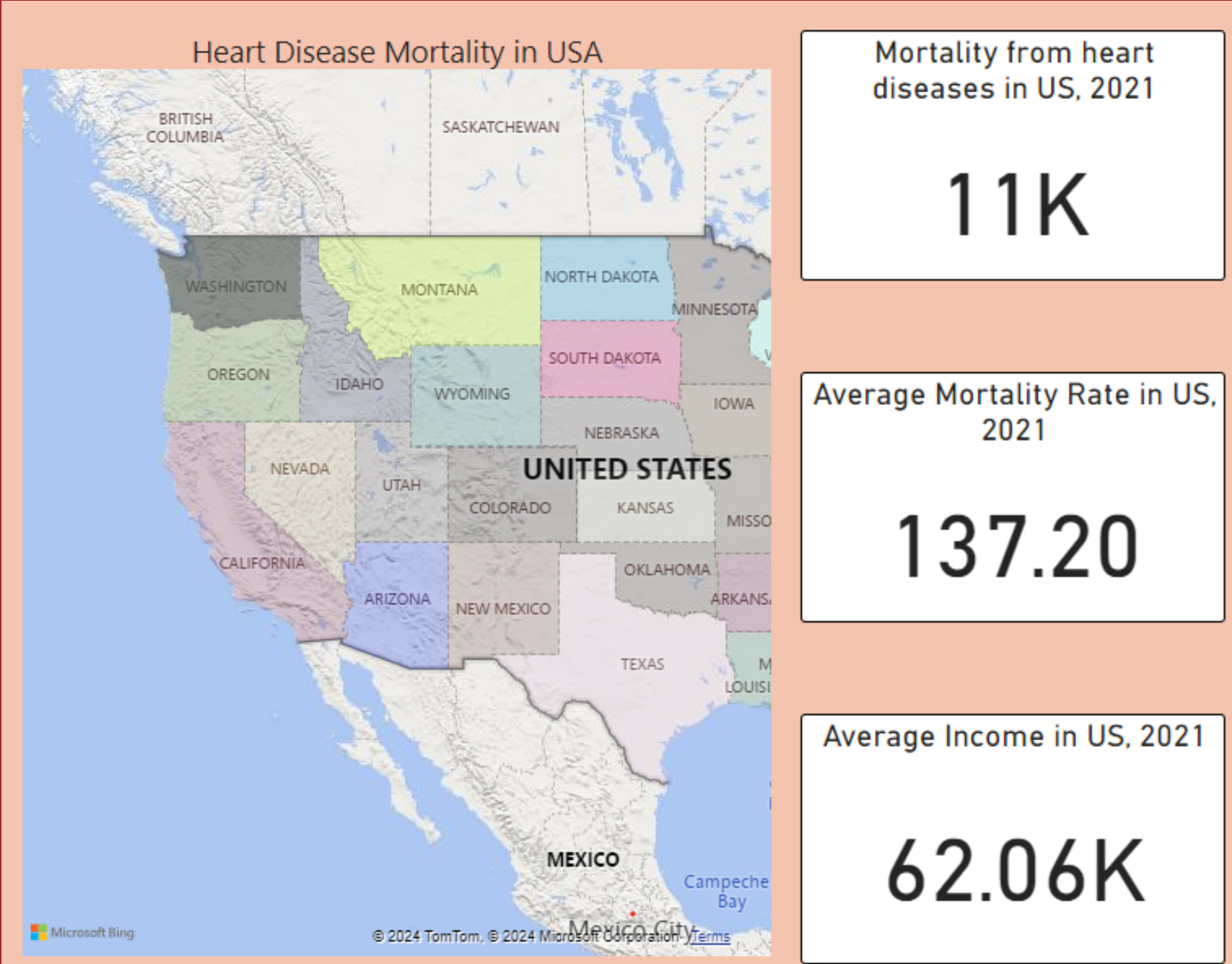
62.06K

VISUALS: US GENERAL

Visuals used:

- Table
- Filled Map
- Cards
- Button
- Bookmark

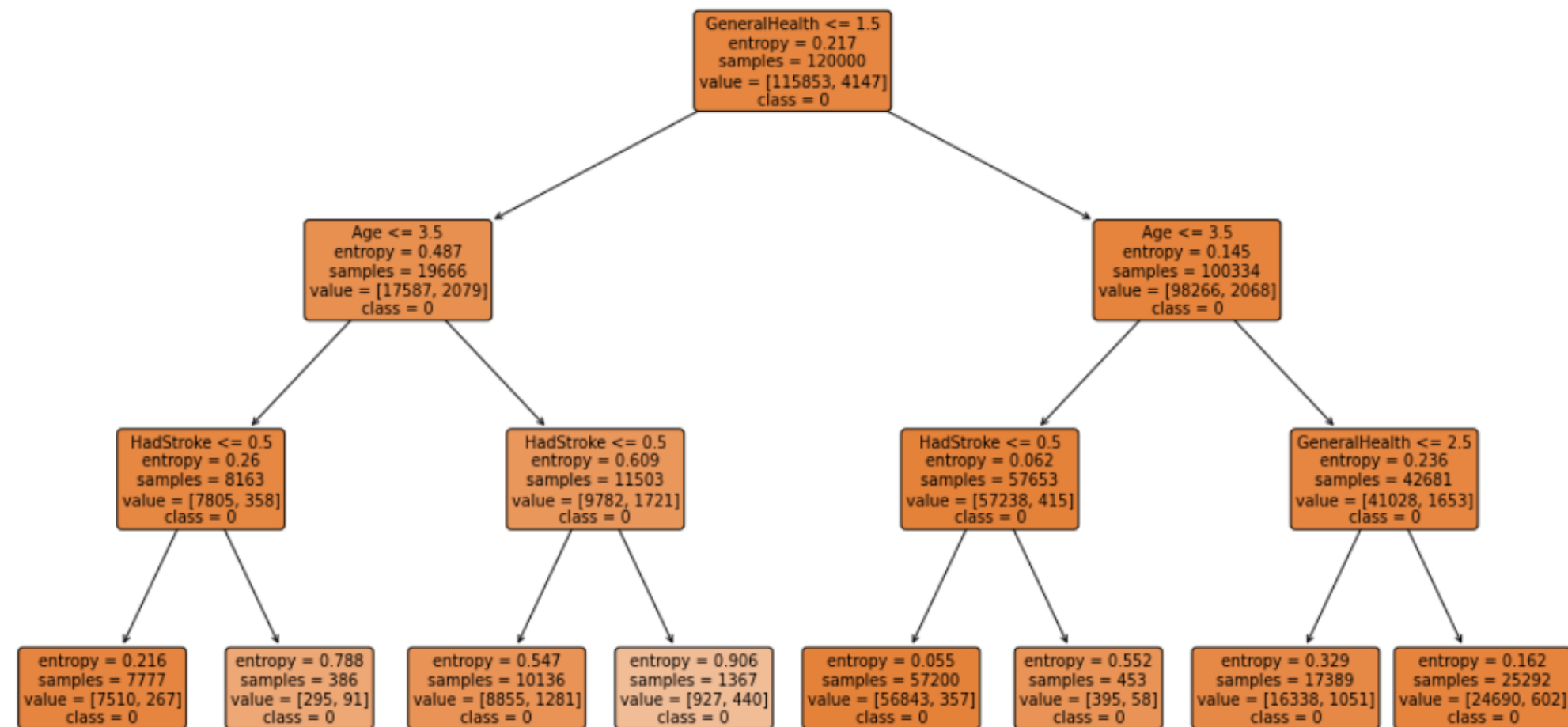
State	AvgIncomePerCapita2021	Number of Heart Attacks
Washington	73,755.00	701
Ohio	56,955.00	589
Florida	62,136.00	557



VISUALS: ID3 MODEL

AgeCategory, AlcoholDrinkers, BMI, GeneralHealth, HadDepressiveDisorder, HadDiabetes, HadHeartAttack, HadSkinCancer, HadStroke, MentalHealthDays, PhysicalHealthDays, Sex, SleepHours and SmokerStatus

depth = 3 (or 8)



Q&A

THANK YOU!

REFERENCES

Datasets:

<https://www.kaggle.com/datasets/kamilpytlak/personal-key-indicators-of-heart-disease/data>

https://www.cdc.gov/nchs/pressroom/sosmap/heart_disease_mortality/heart_disease.htm

<https://worldpopulationreview.com/state-rankings/average-income-by-state>

REFERENCES

Other sources:

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8885242/>

<https://www.ncbi.nlm.nih.gov/books/NBK541070/>

[https://www.cnbc.com/2022/08/30/this-is-how-much-you-should-exercise-to-achieve-the-maximum-benefit.html#:~:text=According%20to%20their%20research%2C%20funded,\(21%2D42%20minutes%20daily\)](https://www.cnbc.com/2022/08/30/this-is-how-much-you-should-exercise-to-achieve-the-maximum-benefit.html#:~:text=According%20to%20their%20research%2C%20funded,(21%2D42%20minutes%20daily))

<https://www.verywellmind.com/when-and-how-to-take-a-mental-health-day-3144754#:~:text=Taking%20a%20mental%20health%20day%20often%20means%20taking%20time%20out,relaxing%20once%20they%27re%20completed>